

Human Skeleton Tracking

This program watches a video, finds the people in it, and works out five things about how each person is moving their joint angles, their speed, the path they have walked, how many squats they have done, and which way their body is facing.

How it works (overall)

For every frame the program does three things. First it finds each person and 17 points on their body (such as the shoulders, hips and knees), and gives each person an ID number that stays the same as they move around. Second, it uses those points to work out the five measurements below. Third, it draws the skeleton and the numbers back onto the video. Points the model is unsure about are ignored, and it remembers the last 64 frames for each person so the measurements stay steady.

The five features

Feature	How it works
1. Joint angles	The bend at a joint, in degrees (180 = straight, smaller = more bent). The program takes a joint and the two body parts attached to it and measures the angle between them. It uses whichever side of the body, left or right, is clearly visible. It measures the knee (hip–knee–ankle) and the elbow (shoulder–elbow–wrist). The knee angle is also used by the squat counter but the elbow is only shown.
2. Speed	How fast the person is moving. The program follows the point midway between the hips and remembers where it was over the last 64 frames. Speed is the distance between the oldest and newest position divided by the time between them. To show it in metres per second it assumes the person is about 1.70 m tall, so the value is an estimate rather than an exact measurement.
3. Trajectory (path line)	The recent path the person has walked. The program joins up the last 64 hip-centre positions into a line that fades out towards the older end, with a small dot at the current position. Each person's line has its own color..
4. Squat count	How many squats the person has done. The program watches the knee angle, when the knee bends past 120 degrees it is counted as 'down', and when it straightens past 160 degrees again that counts as one squat. Using two different angles stops it from miscounting when the knee wobbles near a single value.
5. Orientation (body facing)	Which way the body faces, and how much it leans to the side. If the shoulders look close together the person is side-on, and the nose tells left from right. If the shoulders look wide apart the person is facing toward or away from the camera, and a visible face means toward. Lean is how tilted the line from the shoulders to the hips is. It is shown as an arrow and as text.

Problems faced, and how they are handled

- **Body points that are hidden** — if the model is unsure about a point it is ignored.
- **Shaky readings** — speed is averaged over 64 frames instead of comparing one frame to the next.
- **No camera calibration** — metres are guessed from the person's height, so speed is approximate.
- **Miscounting squats** — two different knee angles (120 and 160) keep the count from flickering.
- **People near vs far from the camera** — facing is judged by comparing shoulder width to body height, so it works at any distance.
- **Telling toward from away** — hard from a flat image, so the program also checks whether the face is visible; it can still be wrong when the face is hidden.